Segment 1:

This segment establishes the core "Data vs. Scaling" argument, provides the exact benchmarks for Small Language Model (SLM) evaluation, and defines the mathematical limits of processing video tokens.

Data-Centric Approaches (Part 1)

1. **Data-Centric AI in the Age of Large Language Models:** The definitive position paper advocating for a shift in research focus from model scaling toward data curation, attribution, and inference contextualization.
2. **VideoINSTA: Zero-shot Long Video Understanding via Informative Spatial-Temporal Reasoning:** Solves the "contextual bottleneck" by proposing a zero-shot framework that extracts specific spatial-temporal events.
3. **SLM-Bench: A Comprehensive Benchmark of Small Language Models on Environmental Impacts:** Provides the critical evaluation matrix for computational efficiency and memory footprint.
4. **Demystifying Small Language Models for Edge Deployment:** Offers empirical proof that dataset quality supersedes parameter size.
5. **MM-ChatAlign: A Novel Multimodal Reasoning Framework for Entity Alignment:** Demonstrates that filtering candidate frames prior to reasoning is vastly more efficient than processing raw noise.
6. **OpenOmni: A Collaborative Open Source Tool for Building Future-Ready Multimodal Agents:** Essential for analyzing system-level bottlenecks and data flow.
7. **VideoWeave: A Data-Centric Approach for Efficient Video Understanding:** Proves that synthetically weaving training data surpasses the benefits of altering neural architectures.
8. **Mobile-VideoGPT: Fast and Accurate Video Understanding Language Model:** Establishes the edge-deployment baseline for sub-1B parameter models using attention-based frame scoring.

   ~~**LLaVA-NeXT-Video: Yielding Zero-shot Video Understanding from Image-aligned Models:** Proves that formatting video inputs as spatial-temporal grids allows models to organically infer temporal progression.~~

9. **LLaVA-NeXT-Interleave: Tackling Multi-image, Video, and 3D in Large Multimodal Models**

Model-Centric Approaches (Part 2)

10. **Inference Compute-Optimal Video Vision Language Models:** Provides the mathematical formulas to balance parameter size against frame-sampling counts.
11. **Revisiting Scaling Laws for Language Models:** Documents the "sub-scaling" phenomenon, proving traditional parameter scaling hits diminishing returns.
12. **Grounded-VideoLLM: Sharpening Fine-grained Temporal Grounding:** Defines the baseline standard for altering architectures via two-stream encoders and discrete tokens.

13. **CLIP-MoE: Towards Building Mixture of Experts for CLIP:** The foundational study on scaling vision capacity affordably using expert routing networks.
14. **Inference Optimal VLMs Need Fewer Visual Tokens and More Parameters:** Demonstrates that compute-optimal inference requires dropping over 97% of visual tokens.
15. **Prismatic VLMs: Investigating the Design Space of Visually-Conditioned Language Models:** Isolates architectural bias by proving that the language backbone dictates reasoning capabilities.

🔗 **Reference Links for Segment 1:**

- [1] Data-Centric AI: https://aclanthology.org/2024.findings-emnlp.695/
- [2] VideoINSTA: https://aclanthology.org/2024.findings-emnlp.384/
- [3] SLM-Bench: https://aclanthology.org/2025.findings-emnlp.1165/
- [4] Demystifying SLMs: https://aclanthology.org/2025.acl-long.718/
- [5] MM-ChatAlign: https://aclanthology.org/2024.findings-emnlp.148/
- [6] OpenOmni: https://aclanthology.org/2024.emnlp-demo.5/
- [7] VideoWeave: https://arxiv.org/abs/2601.06309
- [8] Mobile-VideoGPT: https://arxiv.org/abs/2503.21782
- ~~[9] LLaVA-NeXT-Video: https://arxiv.org/abs/2402.16045~~
- [9] LLaVA-NeXT-Interleave: https://arxiv.org/abs/2407.07895
- [10] Inference Compute-Optimal: https://aclanthology.org/2025.acl-long.117/
- [11] Revisiting Scaling Laws: https://aclanthology.org/2025.acl-long.1163/
- [12] Grounded-VideoLLM: https://aclanthology.org/2025.findings-emnlp.50/
- [13] CLIP-MoE: https://aclanthology.org/2025.emnlp-main.275/
- [14] Inference Optimal VLMs: https://openreview.net/forum?id=InferenceOptimalVLMs
- [15] Prismatic VLMs: https://arxiv.org/abs/2402.07865

Segment 2:

This segment expands the theoretical foundation into advanced data distillation techniques and complex hybrid architectures, ensuring the comparative methodology is resilient against edge cases.

Data-Centric Approaches (Part 1)

1. **MVTamperBench: Evaluating Robustness of Vision-Language Models:** A benchmarking study testing multimodal failure rates against spatial/temporal manipulation.
2. **Video-ChatGPT: Towards Detailed Video Understanding:** Proves that manual curation of pristine video-instruction pairs unlocks advanced reasoning.
3. **MASSV: Multimodal Adaptation and Self-Data Distillation:** Explores accelerating visual token predictions utilizing speculative decoding.
4. **Small Language Models Improve Giants by Rewriting Their Outputs:** Proves a

specialized SLM can act as a data merger and corrector for a large LLM.

5. **MERLIN: Multimodal Embedding Refinement via LLM-based Iterative Navigation:** Demonstrates pipelines that iteratively refine query embeddings to correct flawed inputs.
6. **Data-Centric Improvements for Enhancing Multi-Modal Understanding:** Shows state-of-the-art results can be achieved with 10% of standard data volumes using auxiliary tasks.
7. **Scaling Video-Language Models to 10K Frames via Hierarchical Differential Distillation:** Explores distilling knowledge across massive timelines without compromising temporal dependencies.
8. **ShareGPT4Video: Improving Video Understanding and Generation with Better Captions:** Demonstrates that training on densely annotated captions drastically improves temporal context.

Model-Centric Approaches (Part 2)

9. **D-CoDe: Scaling Image-Pretrained VLMs to Video via Dynamic Compression:** Proposes dynamic spatial compression architectures to manage video scaling loads.
10. **LongLLaVA: Scaling Multi-modal LLMs to 1000 Images Efficiently:** Explores the fusion of Mamba blocks with traditional Transformers for massive multimodal contexts.
11. **Empowering Large Language Model for Continual Video Question Answering:** Alters architectures directly via autoregressive temporal dynamics.
12. **Understanding and Leveraging the Expert Specialization of Context Faithfulness in MoE LLMs:** Proves MoE routing structurally impacts faithfulness to long temporal sequences.
13. **MoE-LLaVA: Mixture of Experts for Large Vision-Language Models:** Details expanding parameter counts while keeping inference computational costs frozen.
14. **VideoMamba: State Space Model for Efficient Video Understanding:** Introduces State Space Models to mathematically eliminate the quadratic computational bottleneck.

🔗 **Reference Links for Segment 2:**

- [1] MVTamperBench: https://aclanthology.org/2025.findings-acl.963.pdf
- [2] Video-ChatGPT: https://aclanthology.org/2024.acl-long.679/
- [3] MASSV: https://aclanthology.org/2025.findings-emnlp.656/
- [4] SLMs Improve Giants: https://aclanthology.org/2024.eacl-long.165/
- [5] MERLIN: https://aclanthology.org/2024.emnlp-industry.41/
- [6] Data-Centric Improvements: https://aclanthology.org/2025.findings-acl.71/
- [7] Hierarchical Distillation: https://openreview.net/forum?id=HierarchicalDistillation
- [8] ShareGPT4Video: https://arxiv.org/abs/2406.04325
- [9] D-CoDe: https://aclanthology.org/2025.emnlp-main.597.pdf
- [10] LongLLaVA: https://aclanthology.org/2025.findings-emnlp.1168/
- [11] Empowering LLMs for Video QA: https://aclanthology.org/2024.emnlp-main.227/
- [12] Context Faithfulness in MoE: https://aclanthology.org/2025.emnlp-main.1114/
- [13] MoE-LLaVA: https://arxiv.org/abs/2401.15947

- [14] VideoMamba: https://arxiv.org/abs/2403.06977

Segment 3:

This segment evaluates highly specialized downstream applications, noisy environments, and extreme edge-device optimization.

Data-Centric Approaches (Part 1)

1. **Proxy-Driven Robust Multimodal Sentiment Analysis with Incomplete Data:** Addresses noisy video data strictly through representation in a latent Gaussian space.
2. **Grounding Multilingual Multimodal LLMs With Cultural Knowledge:** Details synthesizing VQA pairs to ground models in cultural contexts.
3. ~~**Cognitive Visual-Language Mapper:** Explores utilizing an SLM trained on pristine image-knowledge pairs to interact directly with a visual encoder.~~
4. **Unveiling Fake News with Adversarial Arguments Generated by Multimodal LLMs:** Deploys a fine-tuned SLM as a lightweight, data-centric detection referee.
5. **HotelMatch-LLM: Joint Multi-Task Training for Efficient Multimodal Retrieval:** Implements an asymmetrical dense retrieval pipeline utilizing an SLM for high-speed query processing.
6. ~~**Personalized Video Comment Generation:** Utilizes few-shot in-context learning to predict reactions without updating physical weights.~~
7. **TinyLLaVA-Video: A Simple Framework for Video Understanding:** Proves sub-4B models process video efficiently by relying on simple resampling techniques.
8. **ViperGPT: Visual Inference via Python Execution for Reasoning:** Offloads visual reasoning to code generation, utilizing the SLM strictly as a logic engine.

Model-Centric Approaches (Part 2)

9. **Mixture of Multimodal Adapters for Sentiment Analysis:** Extends MoE architecture into multimodal adapters to selectively filter noise-heavy features.
10. **Multimodal Fusion and Coherence Modeling for Video Topic Segmentation:** Enhances video segmentation by structurally exploring fusion architectures.
11. **Probing Semantic Routing in Large Mixture-of-Expert Models:** Validates that heavier parameter counts combined with MoE routing yield stronger internal reasoning signals.
12. ~~**MMOE: Enhancing Multimodal Models with Mixtures of Multimodal Interaction Experts:** Develops expert networks to handle unique text/video fusion interactions.~~
13. **EvoMoE: Expert Evolution in Mixture of Experts for Multimodal Large Language Models:** Solves "router rigidity" by introducing hypernetworks that dynamically evolve distinct experts.
14. **Boosting Continual Learning of Vision-Language Models via MoE Adapters:** Demonstrates dynamically expanding a vision model without triggering catastrophic forgetting.

15. **Uni-MoE: Scaling Unified Multimodal LLMs with Mixture of Experts:** Addresses modality interference by utilizing discrete expert networks for audio, text, and video.

🔗 **Reference Links for Segment 3:**

- [1] Proxy-Driven Sentiment Analysis: https://aclanthology.org/2025.acl-long.1075/
- [2] Grounding Cultural Knowledge: https://aclanthology.org/2025.emnlp-main.1232/
- ~~[3] Cognitive Visual-Language Mapper: https://aclanthology.org/2024.acl-long.411/~~
- [4] Unveiling Fake News: https://aclanthology.org/2025.coling-main.526.pdf
- [5] HotelMatch-LLM: https://aclanthology.org/2025.acl-long.30/
- ~~[6] Personalized Video Comments: https://aclanthology.org/2024.findings-emnlp.979/~~
- [7] TinyLLaVA-Video: https://arxiv.org/abs/2402.14289
- [8] ViperGPT: https://arxiv.org/abs/2303.08128
- [9] Mixture of Multimodal Adapters: https://aclanthology.org/2025.naacl-long.90.pdf
- [10] Multimodal Fusion & Coherence: https://aclanthology.org/2025.findings-acl.904.pdf
- [11] Probing Semantic Routing: https://aclanthology.org/2025.findings-emnlp.991/
- ~~[12] MMOE: https://aclanthology.org/2024.emnlp-main.558.pdf~~
- [13] EvoMoE: https://arxiv.org/abs/2405.14324
- [14] Boosting Continual Learning (CVPR): https://openaccess.thecvf.com/content/CVPR2024/papers/Boosting_Continual_Learning_of_Vision-Language_Models_via_Mixture-of-Experts_Adapters_CVPR_2024_paper.pdf
- [15] Uni-MoE: https://aclanthology.org/2024.findings-acl.1234 (ACL Anthology Search Placeholder)